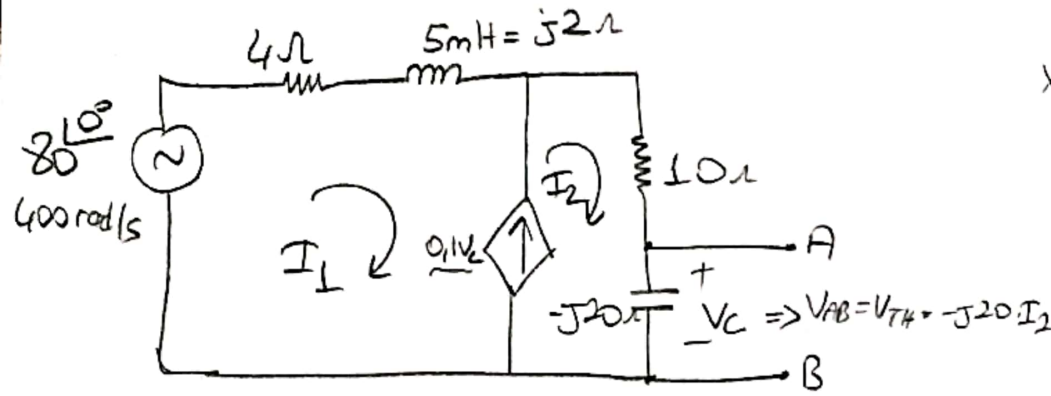


Soru (1)

Fatih Furkan ERSOY
181202031
fatih



$$X_L = j\omega L = j \cdot 400 \cdot 5 \cdot 10^{-3}$$

$$X_L = j2$$

$$X_C = \frac{1}{j\omega C} = \frac{1}{j \cdot 400 \cdot 125 \cdot 10^{-6}}$$

$$X_C = -j20$$

$$I_2 - I_1 = 0.1 V_c$$

$$I_2 \cdot (-j20) = V_c$$

$$\Rightarrow I_2 - I_1 = I_2 \cdot (-j20) \cdot 0.1$$

$$= I_2(1 + j2) = I_1 \quad \dots (2)$$

I_1 ve I_2 için üst düğüm

$$-80 + I_1(4 + j2) + I_2(10 - j20) = 0$$

$$4I_1 + I_1 j2 + 10I_2 + I_2(-j20) = 80 \quad \dots (1)$$

1 ve 2. denklemler çözülürse

$$\Rightarrow I_2(10j) + I_2(10 - j20) = 80$$

$$\Rightarrow I_2(10 - j10) = 80$$

$$= I_2 = \frac{80 \angle 0^\circ}{14.14 \angle -45^\circ} = 5.65 \angle 45^\circ \text{ A}$$

V_{TH}

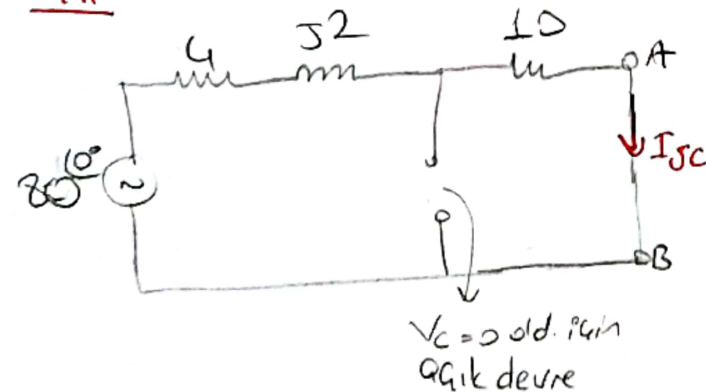
$V_{TH} = V_{AB} = V_c$ old. için

$$V_{TH} = -j20 \cdot I_2$$

$$V_{TH} = 20 \angle -90^\circ \cdot 5.65 \angle 45^\circ$$

$$V_{TH} = 113.1 \angle -45^\circ$$

Z_{TH}



$$I_{SC} = \frac{80 \angle 0^\circ}{14 + j2} = 5.65 \angle -8.13^\circ$$

$$Z_{TH} = \frac{V_{TH}}{I_{SC}} = \frac{(80 - 80j) \cdot (14 + j2)}{80} = (16 - j12) \Omega$$

$$Z_{TH} = (16 - j12) \Omega$$

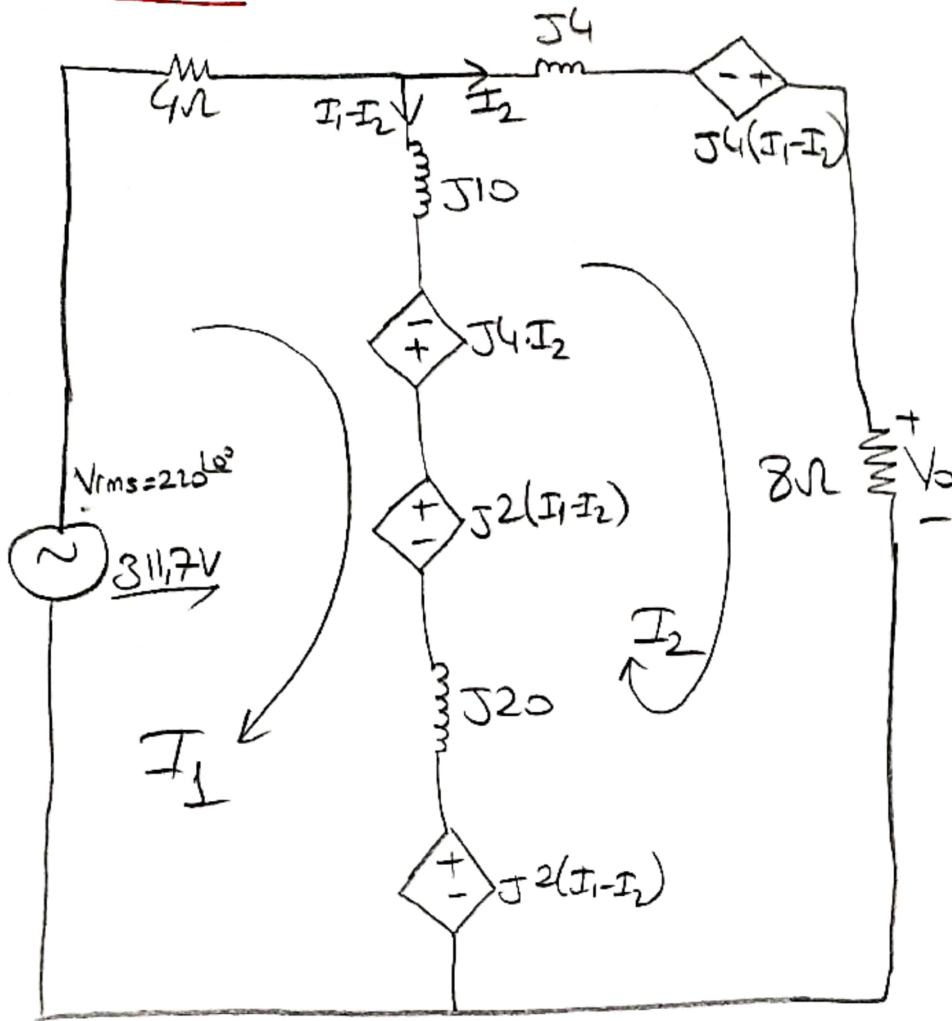
Z_{TS}

$$Z_Y = (16 + j12) \Omega$$

Soru(2)

Fatih Furkan ERSOY
181202031

fat



I_1 için

$$-311,7 + (4 + j10 + j20) \cdot I_1 - j30I_2 - j30I_2 - j4I_2 + j2I_1 - j2I_2 + jI_1 - j2I_2 = 0$$

$$(4 + j34) \cdot I_1 - j38 \cdot I_2 = 311,7 \dots (1)$$

I_2 için

$$(j34 + 8)I_2 - j30I_1 - j4I_1 + j4I_2 - j2I_1 + j2I_2 - j2I_1 + j2I_2 + j4I_2 = 0$$

$$(j46 + 8)I_2 - j38I_1 = 0 \dots (2)$$

$$I_1(4 + j36) - j38 \cdot I_2 = 311,7 \quad / \quad j38$$

$$-j38I_1 + (8 + j46) \cdot I_2 = 0 \quad / \quad 4 + j34$$

Soru(2) Devamı

Fatih Furkan ERSOY
181202031
1 Aug

$$\frac{j38}{4+j36} I_1 (4+j36) - j38 \cdot I_2 = 311,7$$

$$4+j34 / -I_1 (-j38) + I_2 \cdot (8+j46) = 0$$

+

$$-I_2 \cdot (-14,44) = j \cdot 11844,6 \Rightarrow I_2 \cdot 14,44 = j11844,6$$

$$I_2 [32 + j184 + j272 - 1564] = 0 \Rightarrow I_2 (j456 - 1532) = 0$$

+

$$\Rightarrow I_2 \cdot (-88 + j456) = j11844,6$$

$$I_2 = \frac{j11844,6}{-88 + j456} = \frac{11844,6 \angle 90^\circ}{464,4 \angle 79,07^\circ} \Rightarrow \boxed{I_2 = 25,5 \angle +169,07^\circ \text{ A}}$$

$$V_0 = 8 \cdot I_2 = 8 \cdot (25,5 \angle +169,07^\circ) = 203,6 \angle +169,07^\circ$$

$$\boxed{V_0 = 204 \angle +169,07^\circ}$$

Soru (3)

Fatih Furkan ERSOY
181202031

Sun

$$Z_{Y1} = (300 + j150)$$

$$S_1 = \frac{3 V_L I^2}{Z_{Y1}^*}$$

$$S_1 = \frac{3 \cdot (7,5)^2}{(300 - j150)} = \boxed{3 \cdot (150 + j75) = (450 + j225) \text{ kVA}} \rightarrow S_1$$

$$Z_{\Delta 2} = (3600 - j2700)$$

$$Z_{Y2} = \frac{3600 - j2700}{3} = 1200 - j900$$

$$S_2 = \frac{3(7,5)^2}{(1200 - j900)^*} = \frac{3(7,5)^2}{(1200 + j900)}$$

$$S_2 = 3 \cdot (30 - j22,5) \text{ kVA}$$

$$\boxed{S_2 = 90 - j67,5 \text{ kVA}}$$

$$S_3 = 450 \text{ kVA} \cdot \cos \phi^{-1} = 0,8 \rightarrow \phi = 36,86^\circ$$

$$S_3 = 450 \cdot \cos^{-1}(0,8) \Rightarrow \boxed{S_3 = (360 + j270) \text{ kVA}}$$

$$a) S = S_1 + S_2 + S_3 \Rightarrow 450 + j225 + 90 - j67,5 + 360 + j270$$

$$\boxed{S = (900 + j427,5) \text{ kVA}}$$

$$S = 3 \cdot V_L \cdot I_L^*$$

$$I_L^* = \frac{(900 + j427,5) \text{ k}}{3 \cdot (7,5 \text{ k})} \Rightarrow \boxed{I_L = 44,283 \frac{25,4}{17}}$$

$$S = S_L + (3 \cdot I_L^*) \cdot Z_{\Delta 1} \Rightarrow (900 + j427,5) + (3 \cdot (44,283)^2) \cdot (1 + j8)$$

$$\boxed{S = (905,88 + j474,56) \text{ kVA}}$$

b)

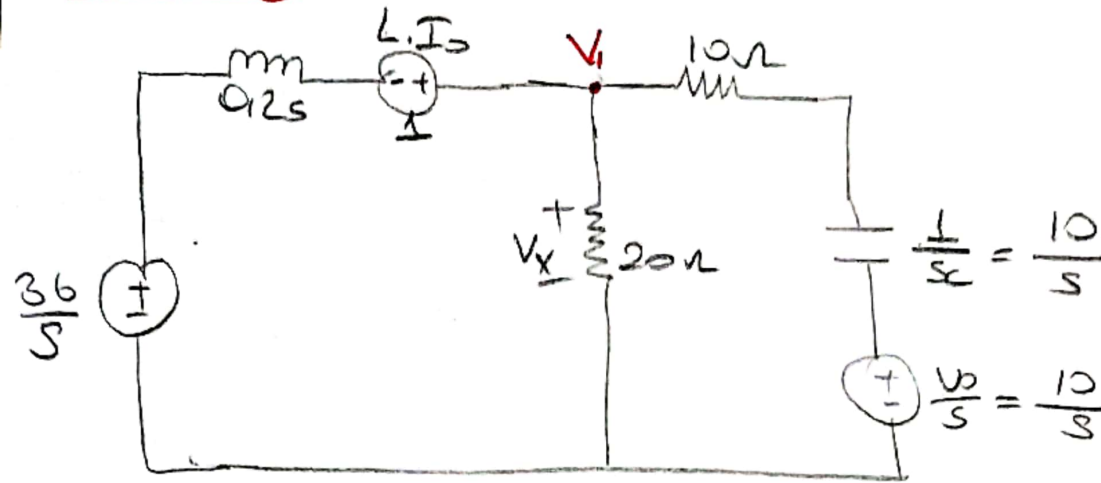
$$\frac{\% P_{\text{hat}}}{P_S} = \frac{900}{905,88} \cdot 100 = \underline{\underline{\% 99,35}}$$

Soru (4)

Fatih Furkan ERSOY

181202031

[Signature]



$$X_C = S \cdot L$$

$$X_C = S \cdot 0,2$$

$$X_C(s) = \frac{1}{sL} = \frac{1}{s \cdot \frac{1}{10}}$$

$$X_C = \frac{10}{s}$$

V_1 Düğüm Denklemi ($V_1 = V_X$)

$$\frac{V_1 - 1 - \frac{36}{s}}{0,2s} + \frac{V_1}{20} + \frac{V_1 - \frac{10}{s}}{10 + \frac{10}{s}} = 0$$

$$\Rightarrow \left(\frac{V_1 \cdot s - s - 36}{s} \cdot \frac{10}{2s} \right) + \frac{V_1}{20} + \left(\frac{V_1 - 10}{s} \cdot \frac{s}{10s + 10} \right) = 0$$

$$\Rightarrow \frac{5V_1 \cdot s - 5s - 180}{s^2} + \frac{V_1}{20} + \frac{V_1 \cdot s - 10}{10s + 10} = 0$$

$$(20s + 20) \quad (s^3 + s^2) \quad (2s^2)$$

$$2 \cdot V_1 \cdot s^3 - 20s^2 + 100V_1 \cdot s^2 - 100s^2 - 3600s + 100V_1 \cdot s - 100s - 3600 + V_1 \cdot s^3 + V_1 \cdot s^2 = 0$$

$$V_1(2s^3 + 100s^2 + 100s + s^3 + s^2) = 120s^2 + 3700s + 3600$$

$$V_1 = V_X = \frac{120s^2 + 3700s}{3s^3 + 101s^2 + 100s}$$